



Physical separation, chemical precipitation, and biological oxidation

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Abstract

The most significant challenge that Egypt is facing nowadays is to Manage and increase sources of clean water. The amount of clean water is decreased in Egypt every day, however, its demand isn't decreased at all. Since this water is needed for agriculture and industry, decreasing the amount of it decreases the agricultural and industrial base, also agricultural and industries bases dump their untreated water in water basins, this leads to the pollution of water, which is a great problem. However, untreated water isn't the only reason for pollution, dumping garbage can also lead to pollution and pollute the water. The solution was selected due to the problems Egypt faces is to treat highly polluted water, using 3 treatment processes; physical: screening of water, chemical: using ALUM $[KAl(SO_4)_2 \cdot 12H_2O]$, and biological: aerated lagoons and UV filter, this treatment will be able to achieve our design requirements which are getting rid of two toxic elements which are Mn(manganese) and Fe(iron). The levels of two toxic elements, iron (Fe) and manganese (Mn), were significantly decreased by the treatment process. The initial concentration of manganese in the water was 1.1mg/L, and it was reduced to 0.3mg/L after the final treatment. Similarly, the initial concentration of iron was 0.68mg/L, and it was also reduced to 0.3mg/L after treatment. A safer and healthier irrigation is provided by the treated water, as the concentrations of these toxic elements are effectively lowered.

Introduction

In Egypt, most water sources close to cities and urban centers are polluted by garbage and dumping of chemicals, legally or illegally. Since the percentage of pollution is increasing, then the percentage of clean water sources is decreasing. Therefore, the industrial and agricultural bases highly depend on consuming clean water. Since Agriculture remains an important sector of the Egyptian economy. Egypt is facing water limitations as its population is growing, the problem was addressed to increase the sources of clean water, increase the agricultural base, and reduce pollution of water (GHONUIM et al., 2019). So, wastewater will be treated by going through three stages (physical, chemical, and biological) to grow a beneficial plant for Egypt.

Our prior solution was Spring Lake ALUM treatment, Spring Lake is a highly eutrophic lake in west Michigan that is impacted by high phosphorus levels. The excessive phosphorus, and other nutrients cause algal blooms in the lake, which degrade its recreational value and ecological health, therefore the use of Alum to treat this lake was a solution. For the advantages: the ALUM increased the quality of the water in the lake and decreased the growth of algae in water. And the Disadvantages are the timeframe for building it was very long. This prior solution benefited us since our prototype was chosen to be screening to remove large solids, adding alum, aerated lagoons and UV lamp was provided to disinfect the water and decrease the growth of microorganisms as in the prior solution the growth of algae decreased (PLSLWD staff, 2018). Our design requirements were chosen to decrease the amount of iron (Fe) and manganese (Mn), because the nervous system, impact reproduction, and damage kidney can be affected by the high levels of Mn and diabetes, hemochromatosis, stomach problems, and nausea are caused by the high levels of Fe. For our solution, the sewage water will pass through screening process, Adding-Alum process, and aerated lagoons accompanied by UV filter. And the amount of Fe was decreased from 0.68 to 0.3mg/L, and the amount of Mn was decreased from 1.1 to 0.3 mg/L.

Materials and methods

Materials

Material	Amount	Usage	Price	Source	Picture
Mechanical timer	1	specific time on the pulleys	95 L.E.	electronics shop	
window grill	1	Remove large solids	7L.E	Metal working shops	
AC motor	2	converts electrical energy into mechanical energy.	40L.E	electronics shop	
plastic containers	3	Holding water	15L.E., 20 L.E	Home goods store	
ALUM	10 gm/L	Aids in the coagulation of negatively charged impurities	5 L.E. for a kilogram	Chemical supply store	
Air pump	1	forcing the oxygen into the water to activate microorganisms	260 L.E	Pet store	
UV filter	1	kills harmful microorganisms in water	240 L.E	Filters shop	
Arugula seeds	14 gm	Our plant to be grown	10 L.E.	Research and information centre	
Pots	5	Where the seeds are planted	3 L.E	Plant supplies store	
Total:755 L.E					

Methods:

The prototype consists of three main parts: a wooden stand was made with a height of 75 cm, a width of 35cm, and a length of 45 cm consisting of two parts 1. Physical Treatment Process: Our first and second processes are held by the first container and it contains: a plastic tap is fixed in this container to move water to the second stage, two motors are connected by two rubber bands and are fixed in the upper shelf at a height of 28cm, a Window grill is Placed between the two motors, that will act like a filtration paper, a mechanical timer is installed and adjusted to run every 15 minutes, Mechanical agitation will be created by the rotation of motors, larger particles and impurities will be trapped by the window wire located between the motors, On the lower shelf, there is a tank that contains our biological treatment 2. Chemical Treatment Process: a solution of Alum is prepared by adding 10 grams of Alum to 1000 ml of water, The solution should be stirred thoroughly until the Alum is completely dissolved, The Alum solution should be added to the water that has been treated by the previous process, The mixing and flocculation process can be enhanced by passing the water through a flocculation machine, The water should be stirred at a fast speed of 200 for two minutes using the flocculation machine. After two minutes, the speed should be reduced to 20 and the stirring should be continued for 15 minutes. The water should be left to settle its impurities for another 15 minutes.3.Biological Treatment Process is divided to two stages: the first stage the aerated lagoons a container is installed a horizontal filter in the air pump is Installed, ensuring it is covered with water. The addition of oxygen to the water will be facilitated by the filter, promoting biological treatment. The filter should be designed to provide aeration and support the growth of beneficial bacteria, which help break down organic matter. The second stage: is the UV filter, the filter is installed and connected to pipes.

Test plan

Water extracted from each treatment can be used for plant irrigation, to test that the Fe and Mn were reduced And we used water analysis to make sure that the amount of toxic elements in the water and the plant decreased. The sample of the treated water was sent to the water treatment station and get the concentration of Fe and Mn. the amount of Fe was decreased from 0.68 to 0.3mg/L, and the amount of Mn was decreased from 1.1 to 0.3 mg/L.In the plant testing To measure the length of the leaves, shoots, and roots. the vernier caliper was used allowing us to measure our desired measurements with high precision every 5 days.

Results

The first (negative trail): is shown in Figure 6

The second attempt (positive trial result): is shown in Table 2

Figure 7,8,9and 10

Table 2
plant grow with different water types

Water type	No. leaves	Length of leaves	Length of shoot	Length of root	Percentage of Fe	Percentage of Mn
City water	4	0.9 cm	2 cm	0.8 cm	0.3 mg/l	0.4 mg/l
Sewage water	2	0.4 cm	1 cm	0.6 cm	0.68 mg/l	1.1 mg/l
1 st treatment	2	0.4 cm	2 cm	1.5 cm	0.44 mg/l	0.85 mg/l
2 nd treatment	2	0.5 cm	1.8 cm	1.4 cm	0.3 mg/l	0.3 mg/l
3 rd treatment	2	0.6 cm	2 cm	1.1 cm	0.3 mg/l	0.3 mg/l

Note: this results was after five days

Figure 7
Length of different parts of the plant

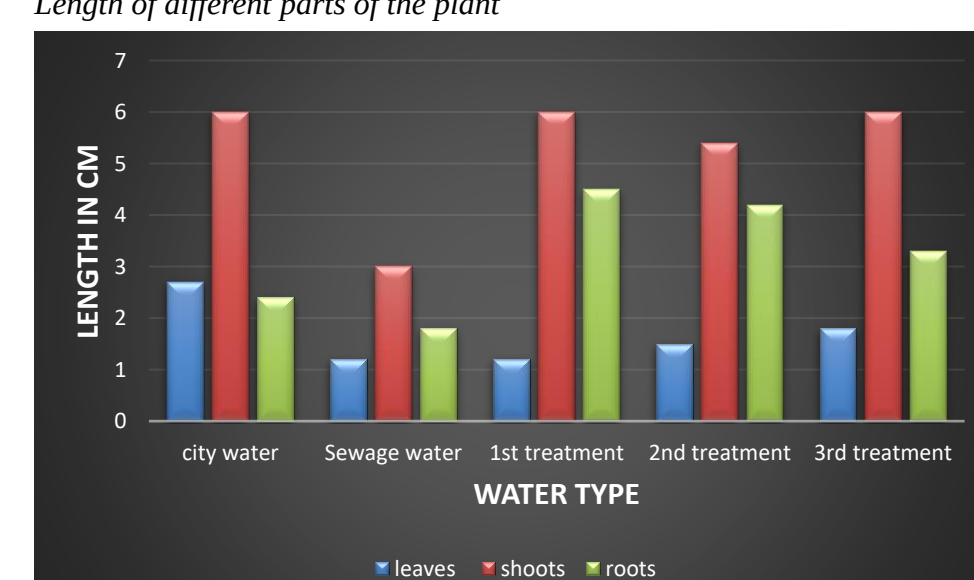


Figure 9
Fe percentage in the different water types

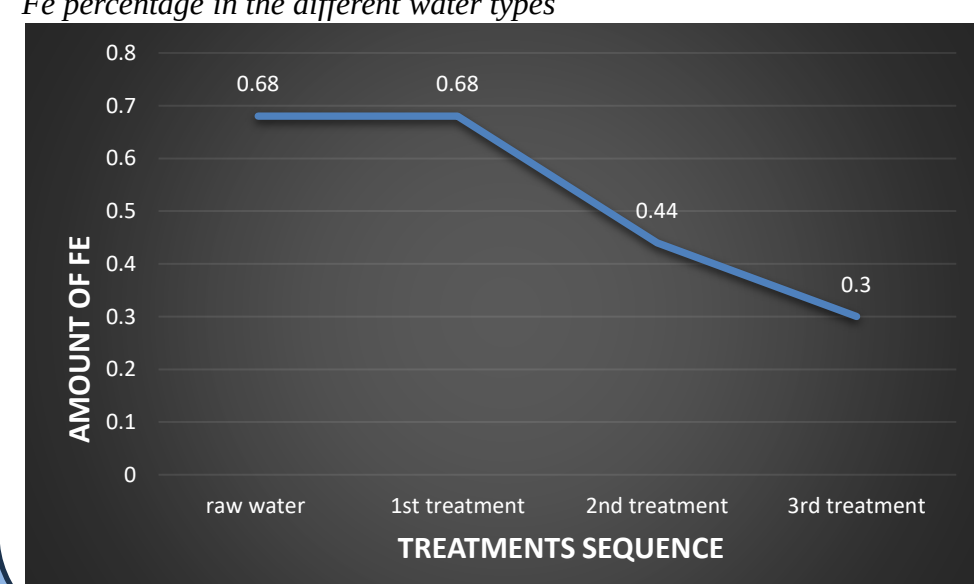


Figure 6
The negative results

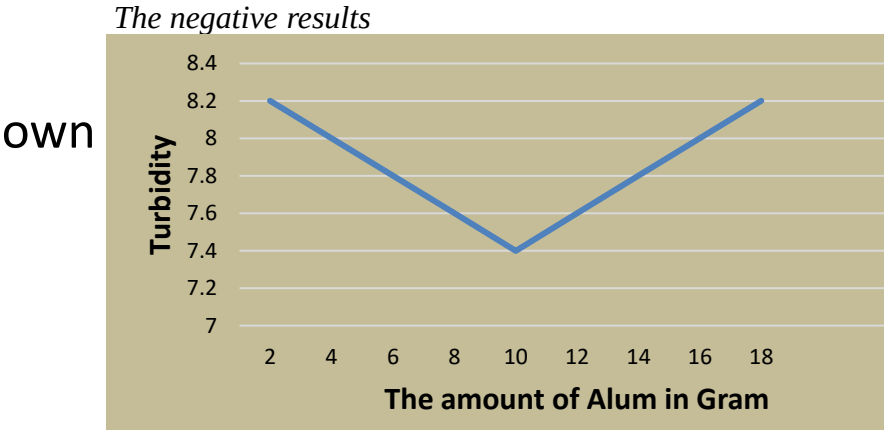


Figure 8
BOD and COD percentage in the different water types

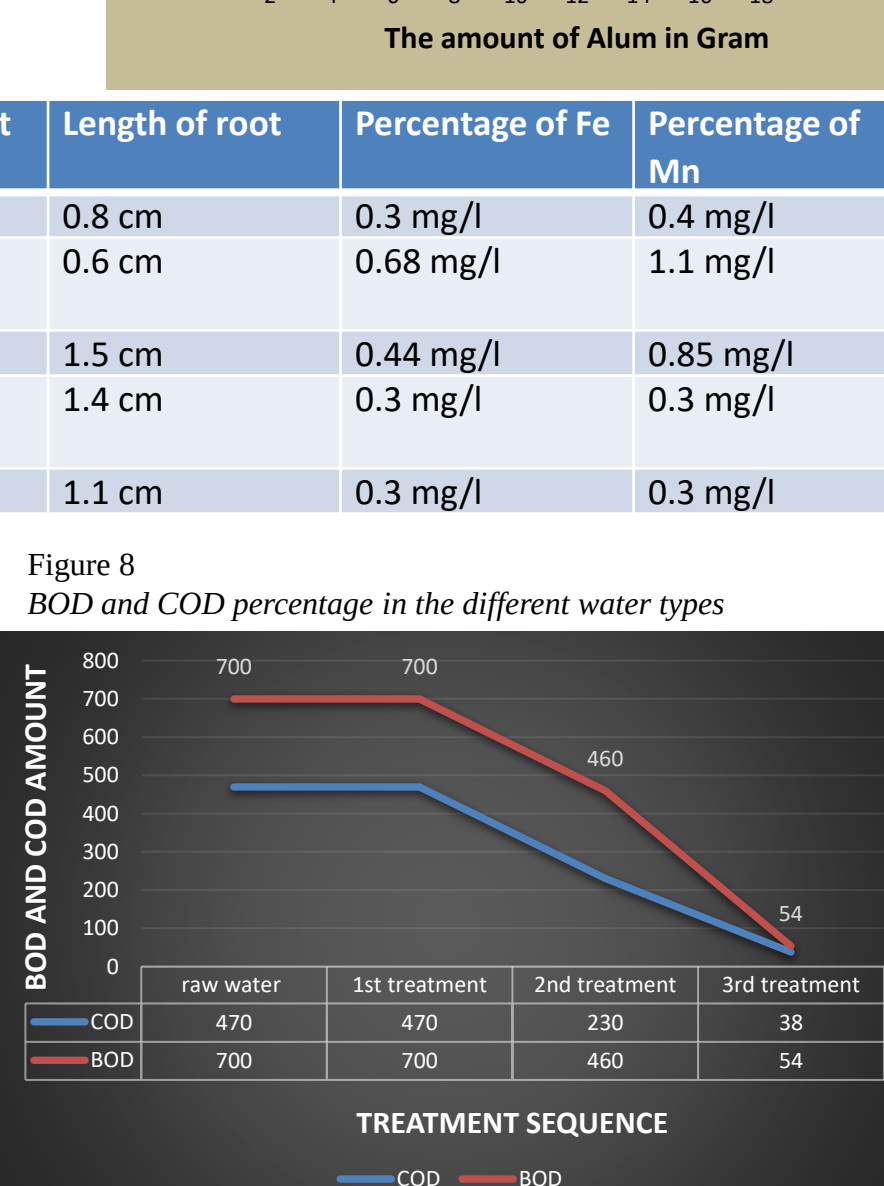
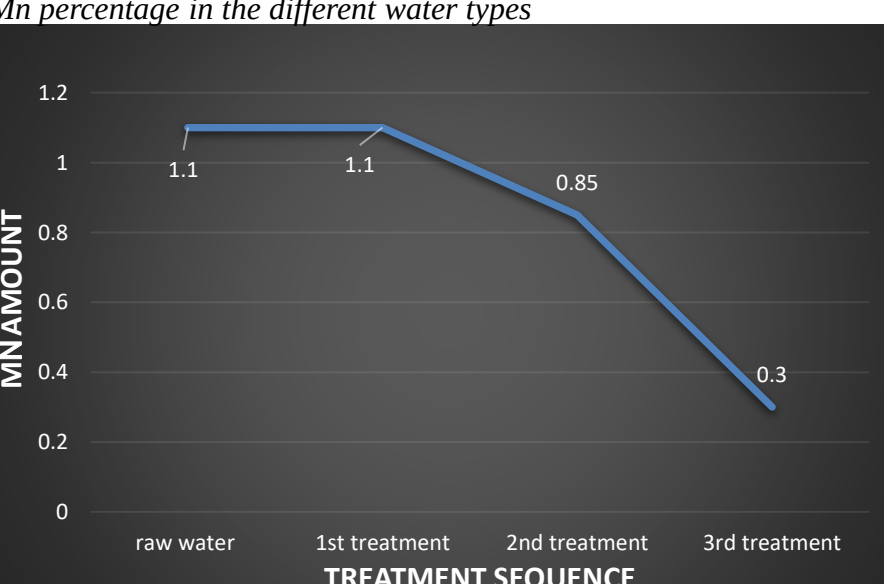


Figure 10
Mn percentage in the different water types



Analysis

The evolution of a community needs the evolution of its problems and ways of thinking; Water treatment is essential for addressing pollution, promoting recycling, supporting agricultural and industrial growth, and increasing water resources. It removes pollutants, conserves freshwater, and provides reliable water supplies for agriculture and industry. Technologies like wastewater treatment expand available water resources. The solution involves using sewage wastewater as a source of untreated water, reducing the demand for freshwater resources and alleviating pressure on limited supplies. In Egypt, where irrigation accounts for 66% of water usage, sewage water is a valuable source (FAO (Food and Agriculture Organization), 2022). Arugula plants were selected for their nutritional value, fast growth, and short harvesting time. The soil composition of 50% clay and 50% sand provides suitable conditions for arugula seed growth, as known in lo 8 geology Texture controls many properties of soil like how fast water drains or can hold or its compaction under heavy loads

Then for the processes, in screening, As known in lo.4 in geology Sewage is first passed through screens. This process removes large pieces of debris and other solids. Solids larger than 6 mm in diameter are removed from the sewage, and then they flow to a system of motors and mechanical timer to make it more efficient. As learned from lo.1 physics water will be attracted to the earth by gravitational force law (Serway & Jewett, 2013)

$$F = G \frac{M_1 M_2}{r^2} \quad g = G \frac{M_e}{r^2}$$

As in ALUM, as we learned in lo.4 geology. To aid in the settling process, alum (hydrated potassium aluminum sulfate), $[KAl(SO_4)_2 \cdot 12H_2O]$ is added. These cause particles of sediment to coagulate. And stick together to form flocs which settle relatively fast, then the sludge is sent to a separate container; the amount of ALUM used will be 10 grams of ALUM in 1000 ml of water, Alum dosage had succeeded in improving the removal efficiency of TDS and Total Iron to 48.15% and 68.11% respectively. In Aerated lagoons As known in lo.1 chemistry the process of dissolving oxygen and we used this concept in our third treatment which is aerated lagoons. Aerated lagoons are a type of wastewater treatment system

That utilizes oxygen to support the biological degradation of organic matter. In these lagoons, dissolved oxygen plays a crucial role in facilitating the treatment process. As known in lo.2 biology oxygen is needed to activate microorganisms such as bacteria and algae, which typically reproduces through a form of cell division called binary fission, which is different from mitosis. In binary fission, a single parent cell divides into two identical daughter cells. The availability of oxygen in water does impact the growth and reproduction of microorganisms, to carry out their metabolic processes. Most microorganisms require oxygen for respiration, which is the process by which they convert organic matter or other energy sources into usable energy. This energy is necessary for cellular functions, growth, and reproduction. (climate policy watcher, 2023). In UV, As we learned in lo.4 geology the organic matter left is digested by microorganisms. Dangerous microbes are disinfected and killed by UV radiation and are guarded against biological pollution during the water movement through pipes. (Nazar & AL-Musawi, 2021)

During the prototype development, the first trial was the negative one encountered a problem when an excessive amount of ALUM was added to the water, causing the formation of an ALUM cloud and turbulence.

The purpose of adding ALUM is to attract pollutants, but an excessive amount has the opposite effect, polluting the water. To address this issue, a smaller percentage of ALUM was chosen which is 10 grams of ALUM for 1000 ml of water, preventing the formation of the cloud and achieving the desired water treatment objective

For the test results: The concentration of manganese in the raw contaminated water was 1.1 mg/l. After the first treatment (screening), it remains constant 1.1 mg/l, and after the second treatment (ALUM), it further decreased to 0.85 mg/l. Subsequently, it decreased to 0.3 mg/l following the third treatment (aerated lagoons) The initial concentration of iron in the contaminated raw water was 0.68 mg/l.

After the first treatment, it is the same .68 mg/l, and following the second treatment, it decreased further to 0.44 mg/l. Subsequently, the concentration decrease to 0.3 mg/l after the third treatment. For the rate of the growth of the plant per five days, a vernier caliper was used to measure the lengths: the length of the roots in the plant irrigated with city water was 0.8 cm, 0.6 cm in sewage water, 1.5 cm in screening, 1.4 cm in ALUM, and finally 1.1cm in aerated lagoons the length of the shoots in the plant irrigated with city water was 2 cm, 1 cm in sewage water, for screening (first treatment) it was 2 cm, for ALUM (second treatment) it was 1.8 cm, and in the aerated lagoons, it was 2 cm. the length and number of leaves respectively, 0.9 cm and 4 respectively for the city, 0.4 and 2 for the sewage, 0.4 cm and 2 for the screening, 0.5 cm and 2 for ALUM, 0.6 cm and 2 for the third treatment, respectively.

Figure 11
Plant with the product water

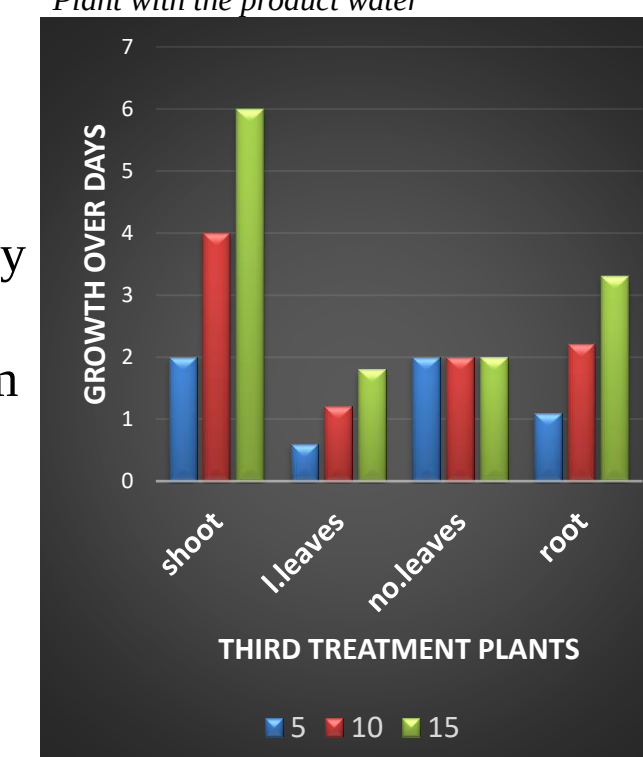


Figure 12
The treatment timeline

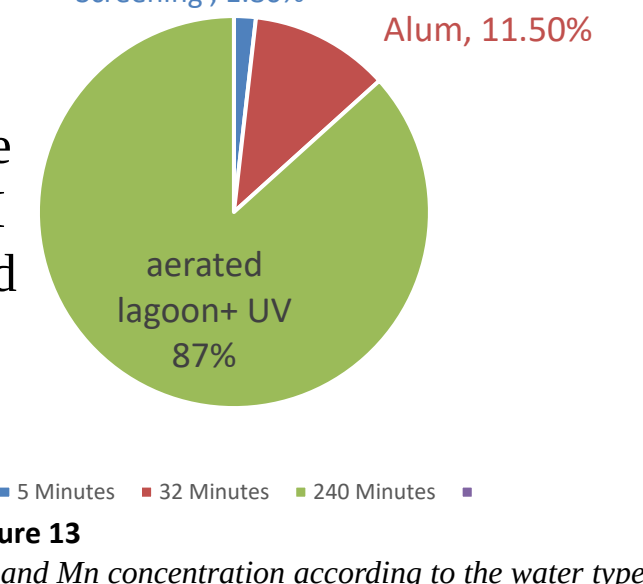


Figure 13
Fe and Mn concentration according to the water type

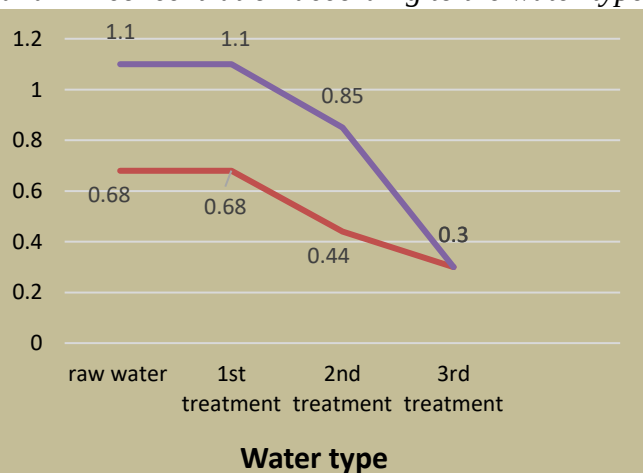
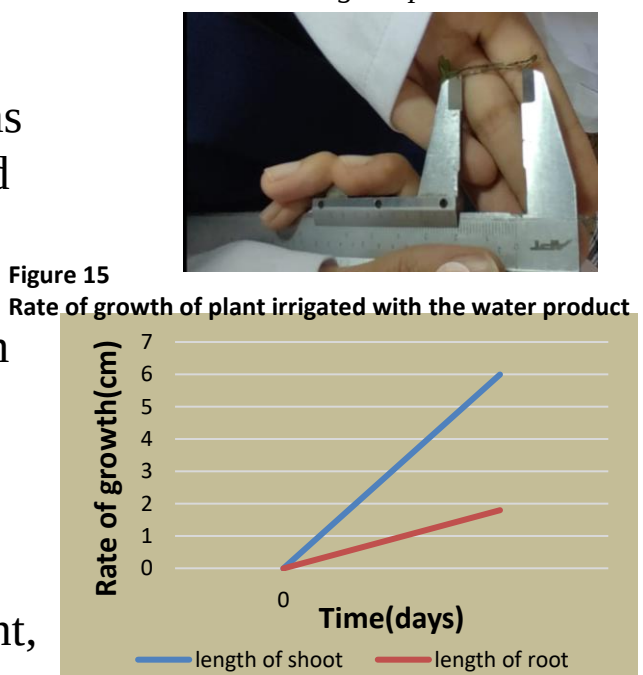


Figure 14
Measuring the plant shoot



Conclusions

After the prototype was constructed, testing was done, and data was collected, some conclusions were made According to the test analysis, the amount of the two toxic elements (Fe and Mn) is reduced in the final treated water as follows: the initial concentration of manganese was 1.1mg/L and after the final treatment it decreased to 0.3mg/L, The initial concentration of iron was 0.68mg/L. and after the final treatment, it decreased to 0.3mg/L. for the plant which was irrigated with the treated water, for the third treatment, the rate of the growth of the plant was: the number of leaves was 2, the length of leaves was 0.6 cm, the length of the shoot was 2 cm, and the length of the root was 1.1 cm, subsequently as the water, and this rate increased by 5 times every five days subsequently, this was achieved by the usage of screening which removes large objects, ALUM which coagulates the contaminants, and aerated lagoons and UV which kill microorganisms and disinfect water; According to the prior solution Spring Lake ALUM treatment, the lake increased the quality of the water in the lake and decreased the growth of algae in the water. As in our solution, the growth of the microorganisms decreases by disinfection by UV light, which increases the quality of water and gets rid of harmful viruses. In addition to reducing the concentration of Fe to 0.3mg/L and the concentration of Mn to 0.3mg/L.

Recommendation

1-Implement an advanced process control system that utilizes real-time data, sensors such as for calculating the amount of toxic elements, and automation to optimize and adjust the treatment processes based on changing influent characteristics that will allow the water to travel between the processes while ensuring the accuracy.

2-Install online monitoring systems for key parameters such as turbidity, pH, dissolved oxygen, ammonia, and phosphorus to enable proactive control and troubleshooting.

3-Utilize data analytics and machine learning techniques to identify patterns, optimize process conditions, and predict potential operational issues or failures.

4- build the water station in a place where there is a water station nearby that can supply with sewage water, and soil which is suitable for Arugula planting.

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